Kunal Vaibhav Thorawade

ee23btech11035@iith.ac.in | thorawadekunal2817@gmail.com | +91 9860057922 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.73
Chhatrapati Shahu Academy and Science Junior College, Satara	Class XII (HSC)	2023	82.67%
Podar International School, Satara	Class X (CBSE)	2021	96.40%

WORK EXPERIENCE

Zinnia SDE Intern:

Spectrum API Service:

- Engineered 50+ API endpoints for MuleSoft decommissioning project, migrating enterprise insurance services to Spring Boot, saving hundreds of thousands of dollars annually by eliminating MuleSoft subscription.
- Designed a multi-system integration architecture to manage complex data flows across independent systems.
- Configured JMS (ActiveMQ) and integrated endpoints with MongoDB, leveraging REST Template and WebClient for API communication.
- Created reusable transformation utilities and standardized response models, reducing code duplication by 50%.

Adaptive Handwritten Data Extraction:

- Engineered prompts for handwritten text extraction, improving structured information extraction from noisy document inputs.
- Designed structured-output prompting strategies for consistent extraction across varied handwriting and document formats.

SKILLS

Programming Languages: C/C++, Java, JavaScript.

Softwares and Tools: Latex, MATLAB, Git/Github, Verilog, LTSpice, Solidedge, Docker.

Frameworks: React, Vite, Spring-Boot.

Integrated Development Environments: Visual Studio Code, Eclipse, Intelliji, Google Colab.

Concepts: Operating Systems (OS), Database Management Systems (DBMS), Object-Oriented Programming (OOPs).

Soft Skills: Leadership, Team Management, Decision Making Skills, Negotiation.

PROJECTS

Automated Codebase Structure & Function Call Analyzer: (Zinnovate 2026)

- Built an automated code analysis tool in a team of 2 during Zinnia's 48-hour hackathon to clone and map Java GitHub repositories.
- Created a static analysis engine using Java (Spring Boot) and C++ Tree-Sitter to extract files, classes, and method calls without compiling the code.
- Mapped code structures into a Neo4j graph database using batch processing to save thousands of code relationships.

Expense Splitting & Debt Settlement Engine:

- Built an O(N log N) debt-simplification algorithm using min-max heaps to settle complex group balances with the absolute fewest payments possible.
- Designed a financial ledger in MySQL paired with Redis caching to calculate user balances accurately without any decimal rounding errors.

- Developed REST APIs in Spring Boot with JWT authentication and containerized the system using Docker Compose.

Real Estate Management System:

- Implemented a secure multi-role user system with modular RESTful endpoints covering authentication, property management, lease tracking, payments, and notifications.
- Interactive Real Estate Management System built using MySQL database in 3NF for integrity and efficiency.
- Leveraged stored procedures, scheduled events, and triggers in MySQL to automate workflows.

POSITIONS OF RESPONSIBILITY

Production & Pronites Head - Elan & nVision IIT Hyderabad: Led a team of 5 coordinators to plan and execute Elan & nVision 2026 proshows, managing artist outreach, negotiations, logistics, and concert operations for artists including Mohit Chauhan.

Partnerships & Collaborations Coordinator - TEDxIITHyderabad: Pitched TEDxIITHyderabad to companies and secured sponsorships from PETA India, Carrier India, and Lotte while building long-term partnerships for future events.

Placement Coordinator - OCS IIT Hyderabad: Actively pitiching to companies to take part in placement drive and managing them.

UG Mentor - Sunshine IIT Hyderabad: Was part of the College Counselling Club in 2nd year and conducted various fun and interactive sessions for freshers.

EXTRACURRICULAR

- Secured 3rd place among 40 teams in Zinnia's annual internal hackathon, Zinnovate.
- Won 3rd prize in the Mad Over Marketing competition at Milan.
- NSS volunteer involved in social service activities.

RELEVANT COURSES

Computer Science Fundamentals: Introduction to Programming, Introduction to AI/ML, Data Structures and Algorithms.

Mathematics: Matrix Theory, Vector Calculus, Probability and Random Processes.

Finance & Management courses: Finance for Engineers, Fundamentals of Organizational Structure.